Formal Language and Automata Theory

FLAT

Topics Covered

1. Introduction to flat
2. Applications of flat
3. Real world example
4. Importance of flat

Introduction to flat

- FLAT is a theortical branch of computer science that deals with:
- -Formal languages
- -Grammer
- -Automata(machines)
- it studies how machines compute and recognize parrerns

Types of Automata

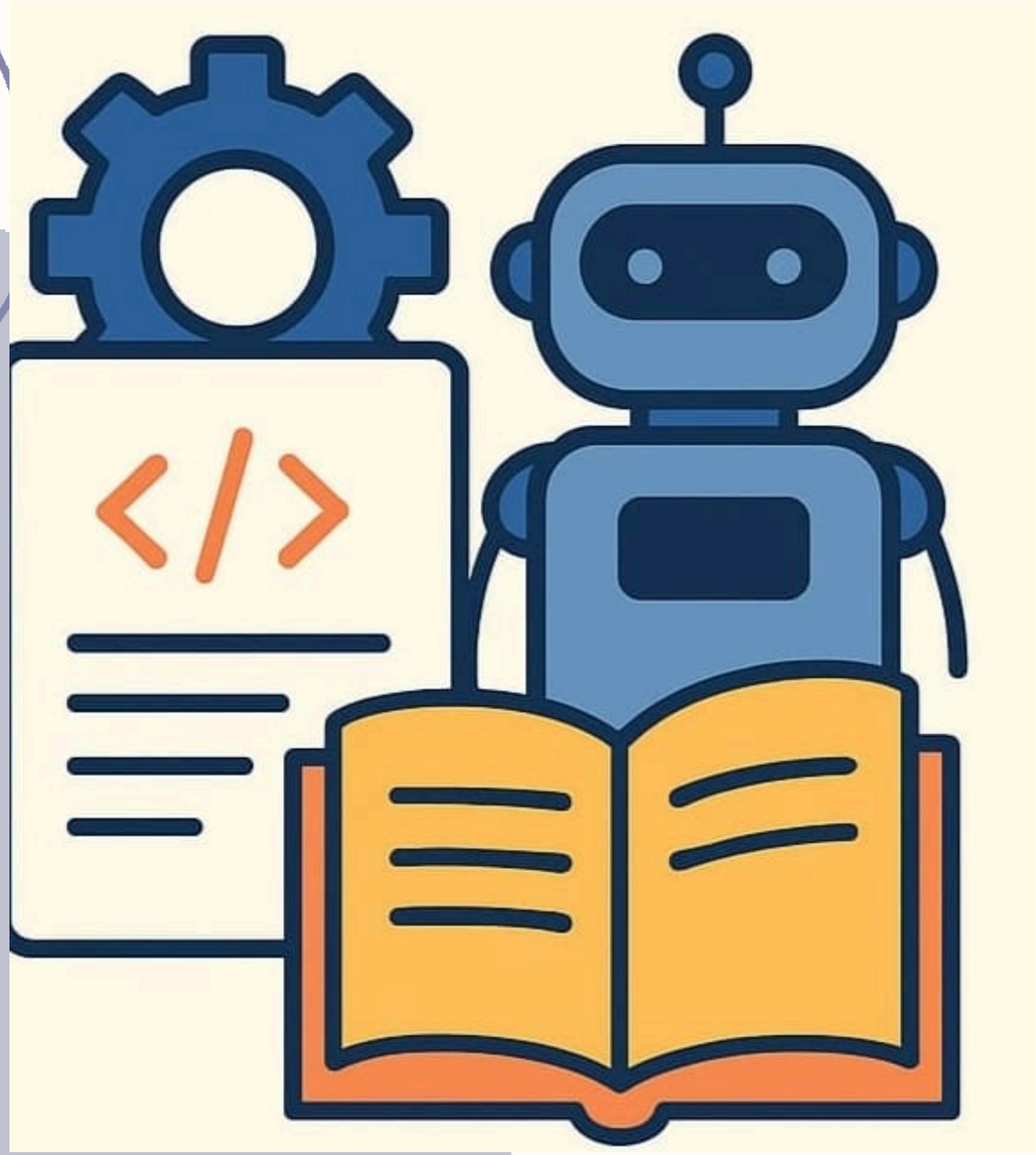
- Finite Automata
- pushdown automata
- turning machines
- each has increasing computational power

Applications of flat

- compiler design
- Artifical intelligence
- Cyber security
- Digital circuit design

Real world example

- Search Engines: pattern matching
- Lexical Analyzers: compilers
- Text Editor: syntax highlighting
- AI: natural language generation



Importance of flat

- Helps in designing parser and compliers
- Flat is essential in understanding how compilers work
- Improves problem solving
- Applications in security &protocols

Flat in daily life

1. Google search and text predictions
2. Programming languages
3. Voice assistants(like Alexa,siri)
4. Chatbots and AI assistants

Thank you!